

PART A – Conceptual Understanding (Theory)

Q1. What is Logistic Regression and why is it suitable for classification?

Answer:

Logistic Regression is a supervised machine learning algorithm used to classify data into two or more classes using a sigmoid function. It is suitable for classification because it predicts the probability of class membership, such as High Risk or Low Risk.

Q2. Explain classification performance metrics and why accuracy alone is insufficient.

Answer:

Classification metrics include Accuracy, Precision, Recall, F1-Score, and ROC-AUC to evaluate model performance. Accuracy alone is insufficient for imbalanced datasets because it can be high even when the model fails to identify minority-class instances.

Q3. Define Type-I Error and Type-II Error in the context of risk prediction.

Answer:

Type-I Error (False Positive) occurs when a low-risk customer is incorrectly classified as high risk. **Type-II Error (False Negative)** occurs when a high-risk customer is incorrectly classified as low risk.

Q4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Answer:

Precision measures the correctness of positive predictions, while Recall (TPR) measures how many actual positive cases are correctly identified. F1-Score balances Precision and Recall, whereas FPR measures the proportion of negative cases incorrectly classified as positive.

Q5. What is AUC-ROC and how does it help in evaluating classifiers?

Answer:

The ROC curve shows the relationship between the True Positive Rate and False Positive Rate at different thresholds. AUC measures the model's ability to distinguish between classes, where a higher value indicates better performance.

Q6. Why does imbalanced data create problems in classification models?

Answer:

Imbalanced data causes the model to favor the majority class, reducing its ability to detect minority-class instances accurately. Techniques such as SMOTE, ADASYN, and resampling help improve classification performance.